Problem Set 2 — Hypothesis Testing

MGT 407E Fall 2026

Forming hypotheses, t-tests, and p-values

Logistics and Reminders

- Create a PDF of your answers. The easiest path is to add your answers to this document and save it as a PDF, then upload it to the matching assignment on Canvas. Name your file `ps2_lastname_firstname.pdf`. Please feel free to handwrite all or some answers if that is easier. If you do that, just snap a picture of the answer and upload the image to your document before making the pdf.
- Please DO NOT waste time spinning your wheels. If you get stuck please reach out to me, the TA, or a classmate for help. For example, there is no need to waste 30 minutes trying to figure out how to make a pdf, or get an AI to do something.
- Use of AI is permitted *only* where a problem is tagged **AI Allowed** or **AI Required**. Problems tagged **NO AI** must be done by hand (you may still use AI for the routine summary statistics noted in the problem). On **NO AI** problems a plain calculator or spreadsheet for arithmetic is always fine — the ban is on outsourcing the *reasoning*, not on doing the multiplication. Where AI is used, paste a clean record of your final prompts and the AI's output into your answer. Answers without supporting work receive no credit.
- **Checking bonus (+2, on top of the 100-point scale).** Once per submission, you can earn 2 extra points for two checks *you* initiate of an AI output: recomputing one of its numbers yourself, cross-checking it against an outside source, catching and correcting an error in what it produced, or something else you dream up. The check and its result must be visible in what you submit (a separately uploaded companion file, referenced from your answer, counts). Any checks required as part of a question do not qualify for this bonus.
- You may discuss approaches with your Learning Team, the TA and the professor, but each student must complete and submit their own work.
- Notation and constants, used throughout the course: CI^C denotes a C% confidence interval (e.g., CI^{95}). We use the normal approximation everywhere: $z = 1.96$ for two-sided tests at the 5% level and for a CI^{95}; $z = 1.65$ for one-sided tests at the 5% level; $z = 2.58$ for the 1% level and for a CI^{99}.
- Round each final numeric answer to two decimal places unless told otherwise (where decimals make sense — whole numbers for counts and dollar amounts are fine).
- Some problems ask you to report regression results in a "nice" table, like the one I introduced in slide deck 3. A nice regression table reports the constant and variable estimates along with their standard errors beneath them in columns. The last rows of the columns report the R^2, number of observations, and any other important information for that regression.
- Problem sets submitted after the deadline receive no credit.
- The last two items of this problem set are a **pre-upload checklist** — please look at it before you submit — and a **formula sheet**, the "Vocabulary and Formulas" slides from the class decks collected in one place.

Problem 1. Picking Fund Managers (25 points)

AI Required

Your university's endowment is being pitched by a fund whose star manager "beat the market by almost 12% a year over the past decade — an excess return that is statistically significant at the 5% level." Before the committee votes, you decide to check what "significant" track records might look like in a world where *nobody* has any skill. Toward that end, you will simulate 100 no-skill managers, each with a decade (120 months) of monthly returns. The goal of this problem is to give you a deeper understanding of what a p-value can tell you, and what it can't.

(a) (2 points) In a short sentence, state how many of the 100 no-skill managers you *expect* will look skilled, i.e., have a mean return over the 120 months that is statistically different from 0 at the 5% level. Please give a brief explanation for why this is your prediction.

(b) (2 points) Ask the AI to simulate the performance of 100 *unskilled* fund managers over 10 years (120 months) by drawing each "manager's" 120 monthly returns from a normal distribution with mean 0 and standard deviation 4%. Think of the mean here as the manager's excess return compared to the market. The mean of zero represents the fact that they have no skill — they will match the market on average — but the standard deviation of 4% means they can get lucky or unlucky in any given month. **Ask your AI to save the resulting draws on your computer as a dataset that you can return to in case you get interrupted.**[1]

Note: simulation results vary from draw to draw, so your numbers will not match a classmate's; that variation is part of the lesson.

Note: It's always a good idea to look at your datasets. So, after the AI creates it, open it up on your own, ask the AI to show you the first few rows, or make a few data displays. Make sure it looks correct!

(c) (3 points) For each manager, have the AI compute the t-statistic for testing whether that manager's true mean excess return over their 120 months is zero (please write out or have the AI write out the formula for this t-stat), add those t-stats as another column in the dataset, and plot a histogram of the 100 t-statistics with vertical lines at ± 1.96.[2]

(d) (2 points) How many of your managers "look significant" at the 5% level? Compare that count with the number you predicted in (a). In a sentence, provide an intuition either for why your prediction was correct, or for why you got it wrong.

(e) (2 points) Now single out the best manager, i.e., the one with the largest (most positive) t-statistic. Have the AI report her mean monthly excess return $\bar{x}$ and its standard deviation s. Then, please compute her t-statistic your self **by hand – written or typed** from those two numbers, showing the formula and your plug-ins. Confirm with the AI that your calculation matches the value in the dataset.

[1] Starter prompt if you want one: "Simulate 100 fund managers' returns over 120 months by creating a dataset where each row is a manager, the first column is a manager id (1 to 100), and each subsequent column is a draw from a normal distribution with mean 0 and standard deviation 4%."

[2] Starter prompt if you want one: "For each manager, compute the t-statistic for testing whether his true mean excess return is zero, then plot a histogram of the 100 t-statistics with vertical lines at ± 1.96."

(f) (2 points) Have the AI report the star manager's two-sided p-value. In a sentence: if a fund had shown you *only this manager's* track record, would a 5% test have concluded she has skill?

(g) (3 points) Every rejection of the null hypothesis in this simulation is a false positive, because no manager has skill by construction. In 2-3 sentences, in your own words, explain how testing many managers at the 5% level spuriously manufactures "stars," and offer a question you recommend be asked whenever someone shows you an impressive track record to flesh this out.

(h) (2 points) A colleague asks the natural follow-up: "so how would we ever find a manager with real skill?" To address this question, consider a 101st manager who genuinely has some skill. That is, let her true mean excess-of-market return be 0.5% per month (about 6% a year), with the same 4% standard deviation. Ask the AI to simulate her draws for a decade, **add her as another row to the saved dataset**, report her t-stat and the p-value for that t-stat, and tell you her t-stat rank among all 101 managers.

(i) (2 points) Irrespective of whether her p-value is below 0.05, explain, in a short sentence or two in your own words, whether you think the 5% test would reliably single out a truly skilled manager. Why or why not?

(j) (3 points) Now ask the AI to **create a second dataset** of monthly returns for 6 managers: the five best performers among your 100 no-skill managers from the first decade, and the truly skilled 101^{st} manager. Use no-skill draws for the first 5 (mean 0, standard deviation 4%), and skilled draws for the sixth (mean 0.5%, standard deviation 4%). Ask the AI to create a scatterplot of the 6 managers' t-stats in the first decade against their t-stats in the second decade. Add dashed horizontal and vertical lines at ± 1.96 and a dashed 45 degree line to make the plot easier to read.[3] Ask the AI to use a green star to mark the skilled manager, and red dots for the other 5.

(k) (2 points) Ask the AI to pool the skilled manager's two decades (240 months) and recompute her t-stat. In 2-3 sentences, in your own words, explain to your colleague what separates skill from luck, what you might advise the board to do before selecting a new manager to address the problem you raised in part (g) — and, coming full circle, what you recommend the committee do about the star manager whose "significant at the 5% level" track record opened this problem.

Your Deliverable. The histogram with the ± 1.96 lines (c) and the count of "significant" managers compared with your (a) prediction (d); your by-hand computation of the star manager's t, formula and plug-ins shown (e), with her p-value and your verdict (f); your written explanation (g); the skilled manager's decade-one t, p-value, and rank (h), with your take on whether the test finds her reliably (i); the decade-1-vs-decade-2 scatterplot for all six managers (j); and the pooled t with your conclusion about finding genuine skill (k).

[3] If a manager is on the 45 degree line, it means they have the same t-stat in both decades.

Problem 2. Home-Field Advantage on Trial: The Empty-Stadium Experiment

(25 points)

AI Required

Sports teams, leagues, and bettors have argued for a century about *why* home teams win more often. The crowd? Travel fatigue? Referees? In 2020, COVID accidentally ran the experiment: the German Bundesliga played the first 224 matches of its 2019–20 season in front of fans, then finished the final 82 matches in *empty stadiums* ("ghost games"). The file `data_bundesliga_ghost_raw.csv` contains all 306 matches: `date`, `hometeam`, `awayteam`, `homegoals`, `awaygoals`, and `ghost` (= 1 for the empty-stadium matches). The goal of this problem is to increase your understanding of one- vs. two-sided tests of a difference in means. It also introduces you to the idea of a "natural" experiment that we will talk about during our third meeting.

(a) (ungraded) Drop the dataset into the AI and ask it to show you some summary statistics to familiarize yourself with the variables. (You don't have to submit anything for this part — it's just good practice.)

(b) (4 points) Ask the AI to create the variable `margin` = `homegoals` − `awaygoals`. Letting μ_{fans} and μ_{ghost} denote the true mean margins, write out the H_0 and H_1 for a *one-sided* test that removing the fans *reduced* the home team's edge.

(c) (3 points) Ask the AI to create a table like the blank one below that reports each group's n, mean margin, the standard deviation of the margin, and each group's share of home wins (matches with `margin` > 0). Also have the AI plot histograms of `margin` for the fans and ghost matches on the same figure (make sure it is readable; ask the AI to change it in some way if not), marking each group's mean. Ask the AI to put the table and the histogram next to each other to make it easier to paste into your answers.

	n	Mean margin	Std. dev. of margin	Share of home wins
With fans (`ghost` = 0)				
Ghost games (`ghost` = 1)				

(d) (2 points) In 1 sentence, in your own words, describe how the ghost games compare to the regular games. Please indicate whether the outcome surprised you or not, and why?

(e) (7 points) Using the values *exactly as printed in your table* — so that everyone's arithmetic lands on the same numbers — compute the difference in mean margins, its standard error (do not assume equal variance in the two groups), and the associated t-statistic. Ask the AI to give you the *one-sided* p-value for that t-statistic. For the SE and t-stat, please write (or type) out the formulas yourself.

(f) (4 points) In a short sentence, state your conclusion about ghost games in "stat speak", i.e., whether you reject or fail to reject H_0 at the 5% level. In a second sentence, in your own words, state your conclusion in plain English.

(g) (2 points) A skeptical colleague objects: "you should have run a *two-sided* test — maybe ghost games could have helped home teams." Compute the two-sided p-value at the same 5% level. In a sentence: does your conclusion change?

(h) (3 points) In a short sentence or two, in your own words, explain why the verdict is so fragile here, and why the sidedness of a test must be chosen *before* looking at the results.

Your Deliverable. Your hypotheses (b), the completed table and overlaid margin histograms (c), your one-sentence description of how the ghost games compare (d), the difference in mean margins with its standard error and t-stat (formulas shown) and the one-sided p-value (e), your conclusion in stat speak and plain English (f), the two-sided p-value and whether the verdict changes (g), and your explanation of the fragile verdict (h).

Problem 3. Audit the AI: The Data Behind the Test (25 points)

AI Required

You are the COO of a food-delivery startup whose marquee promise is "**30 minutes, on average**." For the board meeting, your operations team hastily pasted together `data_deliveries_raw.csv` (`orderid`, `zone`, `minutes`) from several zone spreadsheets, a re-exported "late deliveries" investigation file, and the order log. A board member dropped the file into an AI assistant. The member's prompt, and the reply are in the following box.

AI session transcript

Prompt: Here is our delivery data. Test whether our average delivery time meets our 30-minute promise.

AI: I tested $H_0 : \mu = 30$ against $H_1 : \mu \neq 30$. The file contains $n = 498$ deliveries with mean 94.9 minutes and standard deviation 341.4, so $\text{SE} = 341.4/\sqrt{498} = 15.3$ and

$$t = \frac{94.9 - 30}{15.3} = 4.24, \qquad p < 0.0001.$$

We reject the null hypothesis: average delivery time is *more than triple* the promise. I recommend an immediate overhaul of delivery operations.

The AI's arithmetic is flawless. Its conclusion is garbage. The goal of this problem is to drive home the importance of examining your data before drawing conclusions from them. (The AI may do the computing throughout; the *diagnosis* — spotting what is wrong and deciding the cleaning rules — must be your own.)

(a) (9 points) Do what Slide Deck 2 tells you to do before trusting any AI analysis: have the AI summarize the data and plot a histogram of `minutes`.[4] There are **three** distinct problems hiding in the file. Identify all three, citing the evidence for each.

(b) (3 points) Propose a solution for each problem, and then implement these solutions on a second copy of your dataset.

(c) (4 points) Re-run the AI's test of $H_0 : \mu = 30$ on your cleaned data (the AI may do the computing, but show every piece: your cleaned n — report it explicitly — the mean, sd, SE, and t), and give the p-value and your verdict at the 5% level.

(d) (3 points) Construct the CI^{95} for the true mean delivery time. A CI^{95} and a 5% two-sided test should always agree. Show that this is the case here, i.e., that the test fails to reject $H_0\colon \mu = 30$ exactly when 30 falls inside the interval.

(e) (6 points) Write a 2-3 sentence memo to the board, in your own words that states your corrected verdict on your startup's 30-minute promise, and the lesson you learned about "statistically significant" results computed on unaudited data.

Your Deliverable. The three problems with evidence (a), your three solutions and their implementation on a copy of the data (b), the corrected test with all pieces shown (c), its CI^{95} with the test–interval agreement confirmed (d), and your memo (e).

[4] Starter prompt if you want one: "Before running any tests on this delivery file: summarize it (n, mean, sd, min, max — overall and by zone) and plot a histogram of `minutes`." What to look *for* is the part that must be yours.

Problem 4. Jumping the Gun (25 points)

AI Required

Your e-commerce growth team is testing a redesigned checkout page. The plan: run the old page (A) and the new page (B) for 30 days, then compare mean revenue per visit to see if they are significantly different. Your eager junior analyst is convinced the new page is better; he watches the dashboard obsessively, recomputing the two-sample t-test every day as the data roll in. He intends to declare victory and adopt the new page the first day the p-value dips below 0.05.

Suppose the true effect of the new page is zero: i.e., revenue per visit on both pages is normal with mean \$100 and standard deviation \$20. The goal of this problem is to see how peeking at results can corrupt a test's conclusions — and how to decide, *in advance*, how long a test needs to run.

(a) (2 points) State the null and alternative hypotheses being tested.

(b) (2 points) Suppose the team had decided to test *once*, at the end of day 30. In a short sentence, state the probability of a false "win" at the 5% level.

(c) (2 points) Your eager analyst will compute a p-value using the cumulative data at the end of every day. That means he will construct 30 p-values. In a second sentence, write down your guess as to whether that procedure will make rejection of the null easier or harder, and why.

(d) (4 points) Assume 40 customers see each page every day, i.e., a total of 80 visitors to the site every day. Ask the AI to simulate the following procedure 500 times:

  (i) Create 30 consecutive days of data, with 40 observations per page per day, drawn from a normal distribution with mean \$100 and standard deviation \$20 — the *same* distribution for both pages, since the true effect of the new page is zero.

  (ii) At the end of each day d, use all of the data up to that point ($40 \times d$ observations per page) and compute the two-sample t-stat and p-value for the difference in the two pages' mean revenue — exactly the test from Problem 2. Each experiment thus produces a sequence of 30 t-stats and p-values, one each per day.

  (iii) Note whether $p < 0.05$ on *any* of the 30 days.

  Have the AI save these draws and calculations in a data file on your computer in case you are interrupted — and make sure the file includes each experiment's full *sequence of 30 daily p-values*, not just the summary shares, because part (g) needs the day-by-day paths. Warn the AI up front that 500 experiments may take a while or time out — running them in batches and accumulating the results is fine, and if the tool keeps struggling, any total of **at least 100 experiments** is acceptable: just report how many you ran (the shares are stable well before 500).

(e) (3 points) Have the AI plot two histograms side by side — each experiment's *lowest* p-value across its 30 days, and its day-30 p-value — and have it write two shares on the figure: the share of your experiments that ever dip below 0.05, and the share with $p < 0.05$ on day 30.

(f) (3 points) Those two shares are likely to be very different. In 1-2 sentences, in your own words, try to explain why. In particular, compare the day-30 share with the false-"win" probability you stated in (b) — do they agree, and should they?

(g) (2 points) Have the AI plot the day-by-day p-value path for one of the sequences of draws in the dataset where the p-value drops below 0.05 in the first 15 days but is above 0.05 on day 30.

(h) (3 points) Using 2-3 sentences, in your own words, explain to the eager junior the problem with "stopping when significant," and what rule should replace it.

(i) (4 points) The chastened junior asks: "fine, one test at the end — but how do we know how long to wait? Is 30 days too little, or too much?" "The answer," you say, "depends on the smallest gain in revenue that would actually change our decision." Suppose your finance group tells you the new page is worth adopting only if it raises revenue by at least \$2 per customer visit. Here is how to work out the answer to the junior's question in three steps:

(i) We know from the formula for standard errors that the SE shrinks as n rises, i.e., as customers accumulate. So, after each page has been seen by n customers, the SE of the difference in mean revenue is $\sqrt{20^2/n + 20^2/n}$ (the formula when each page contributes its own noise). Compute this standard error on day 5 ($n = 5 \times 40 = 200$ per page) and on day 30 ($n = 1{,}200$ per page).

(ii) Recall the test's decision rule: it declares a winner only when the *observed* gap between the two pages is larger than the margin of error, $1.96 \times \text{SE}$ — anything smaller is ruled "could just be noise." If the new page's true gain is \$2, the observed gap will come out *near* \$2, so the test can only detect it once the margin of error has shrunk to \$2 or less. Set $1.96 \times \text{SE} = 2$ and solve for n.

(iii) Convert that n into days at 40 customers per page per day. Was the 30-day plan too short, or more than enough?

Your Deliverable. Your hypotheses (a), your answer to (b), your easier-or-harder guess (c), the side-by-side p-value histograms with both shares (e), your explanation of why they differ, including the comparison with (b) (f), the p-value-path figure (g), your explanation to the junior analyst (h), and your how-long-is-long-enough calculation with its verdict on the 30-day plan (i).

One Last Question (ungraded, but required)

At the end of your answer document, please tell me roughly how much time this problem set took you *in total*, start to finish — reading, AI sessions, write-up, and assembling the PDF. A single number of hours (or a range like "4–5 hours") is perfect. This is not graded and does not affect your score in any way; it helps me calibrate the length of the next problem sets.

Before you upload — a 60-second checklist

On PS1, missing answers and un-checked uploads cost the class more points than any statistical mistake. Please run through this list before you submit:

- Open the PDF you are about to upload — the file itself, not your working document — and read it to the last page. Broken images, unrendered formulas, and missing pages are visible in ten seconds.
- Check that **every lettered part has an answer under it**. Problem 1 runs (a)–(k), Problem 2 (a)–(h), Problem 3 (a)–(e), Problem 4 (a)–(i) — plus the total-time estimate from "One Last Question."
- If any answer says "see separate file," make sure that file is actually attached to the *same* Canvas submission.
- Filename: `ps2_lastname_firstname.pdf`.
- If a table or figure came out different from what you expected (a sign flipped, a share far from your prediction), say so in a sentence rather than leaving it — an unremarked surprise costs points; a remarked one often earns them.

Formula Sheet

The "Vocabulary and Formulas" slides that open the class decks, collected in one place. Notation bridge: the problem sets write s for the sample standard deviation $\hat{\sigma}$, and CI^C (e.g. CI^{95}) for a C% confidence interval.

Populations, samples, and confidence intervals

Population mean	$\mu = \dfrac{\sum_{i=1}^{N} x_i}{N}$
Population variance	$\sigma^2 = \dfrac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$
Population standard deviation	$\sigma = \sqrt{\dfrac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$
Sample mean	$\bar{x} = \dfrac{\sum_{i=1}^{N} x_i}{N}$
Sample variance	$\hat{\sigma}^2 = \dfrac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1}$
Sample standard deviation	$\hat{\sigma} = \sqrt{\dfrac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1}}$
Standard error of $\bar{x}$	$SE(\bar{x}) = \text{Std Err}(\bar{x}) = \sqrt{\hat{\sigma}^2/N} = \hat{\sigma}/\sqrt{N}$
Critical values (CV)	1.65 (90%), 1.96 (95%), 2.58 (99%)
Margin of error of $\bar{x}$ (ME)	$CV \times \text{Std Err}(\bar{x})$
Confidence interval for $\bar{x}$	$\bar{x} \pm \text{Margin of Error}(\bar{x})$

Hypothesis tests and sample size

t-stat (for $\mu = a$)	$t = \dfrac{\bar{x} - a}{\hat{\sigma}/\sqrt{N}}$
t-stat (for $\mu_1 = \mu_2$)	$t = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\hat{\sigma}_1^2/N_1 + \hat{\sigma}_2^2/N_2}}$
Choosing N for a given margin of error (assuming some $\hat{\sigma}$ from a pilot study)	$N = \left(\dfrac{CV \times \hat{\sigma}}{ME}\right)^2$

Covariance, correlation, and the OLS line

Covariance of x and y	$\text{Cov}(y, x) = \sum_i (y_i - \bar{y})(x_i - \bar{x})$
Correlation of x and y	$\rho(y, x) = \dfrac{\text{Cov}(y, x)}{\sigma_y \, \sigma_x}$
OLS slope estimate	$\hat{\beta} = \dfrac{\text{Cov}(y, x)}{\text{Var}(x)}$
OLS intercept estimate	$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$

Regression inference and prediction

Mean squared error (MSE)	$MSE = \dfrac{\sum_i \hat{\epsilon}_i^2}{N-2}$
Standard error of regression	$\text{Std Err of Reg} = \sqrt{\dfrac{\sum_i \hat{\epsilon}_i^2}{N-2}}$
Standard error of $\hat{\beta}$	$\text{Std Err}(\hat{\beta}) = \sqrt{\dfrac{MSE}{\sum_i (x_i - \bar{x})^2}}$
t-stat (for $\beta = a$)	$t = \dfrac{\hat{\beta} - a}{\text{Std Err}(\hat{\beta})}$
Margin of error of $\hat{\beta}$	$CV \times \text{Std Err}(\hat{\beta})$
Confidence interval for $\hat{\beta}$	$\hat{\beta} \pm \text{Margin of Error}$
Regression goodness of fit (R^2)	$R^2 = \dfrac{\text{Var}(\hat{y})}{\text{Var}(y)} = 1 - \dfrac{\sum_i \hat{\epsilon}_i^2}{\sum_i (y_i - \bar{y})^2}$
Predicting y at x^*	$\hat{y} = \hat{\alpha} + \hat{\beta} x^*$
Confidence interval for predicted $\hat{y}$	$CI(\hat{y}) = \hat{y} \pm CV \sqrt{MSE \left(\dfrac{1}{n} + \dfrac{(x^* - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \right)}$
Prediction interval for a new y (optional)	$PI(\hat{y}) = \hat{y} \pm CV \sqrt{MSE \left(1 + \dfrac{1}{n} + \dfrac{(x^* - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \right)}$

Variance rules (assume $\text{Var}(\epsilon_i) = \sigma^2$ and a is a constant)

Scaling by a	$\text{Var}(a\,\epsilon_i) = a^2\sigma^2$, $\quad \text{Var}(\epsilon_i/a) = \sigma^2/a^2$
Sum of n independent ϵ_i	$\text{Var}(\sum_{i=1}^n \epsilon_i) = n\sigma^2$